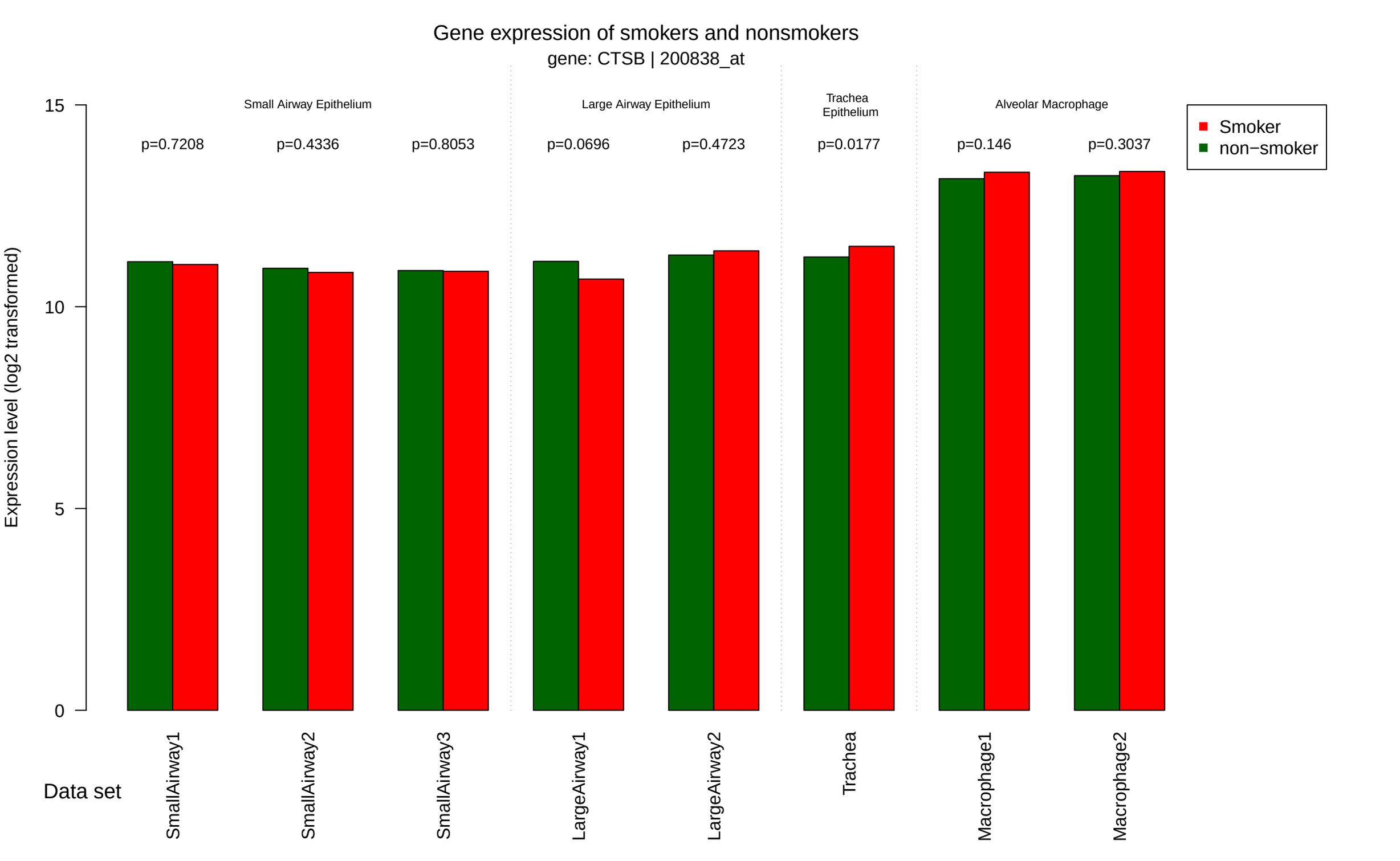
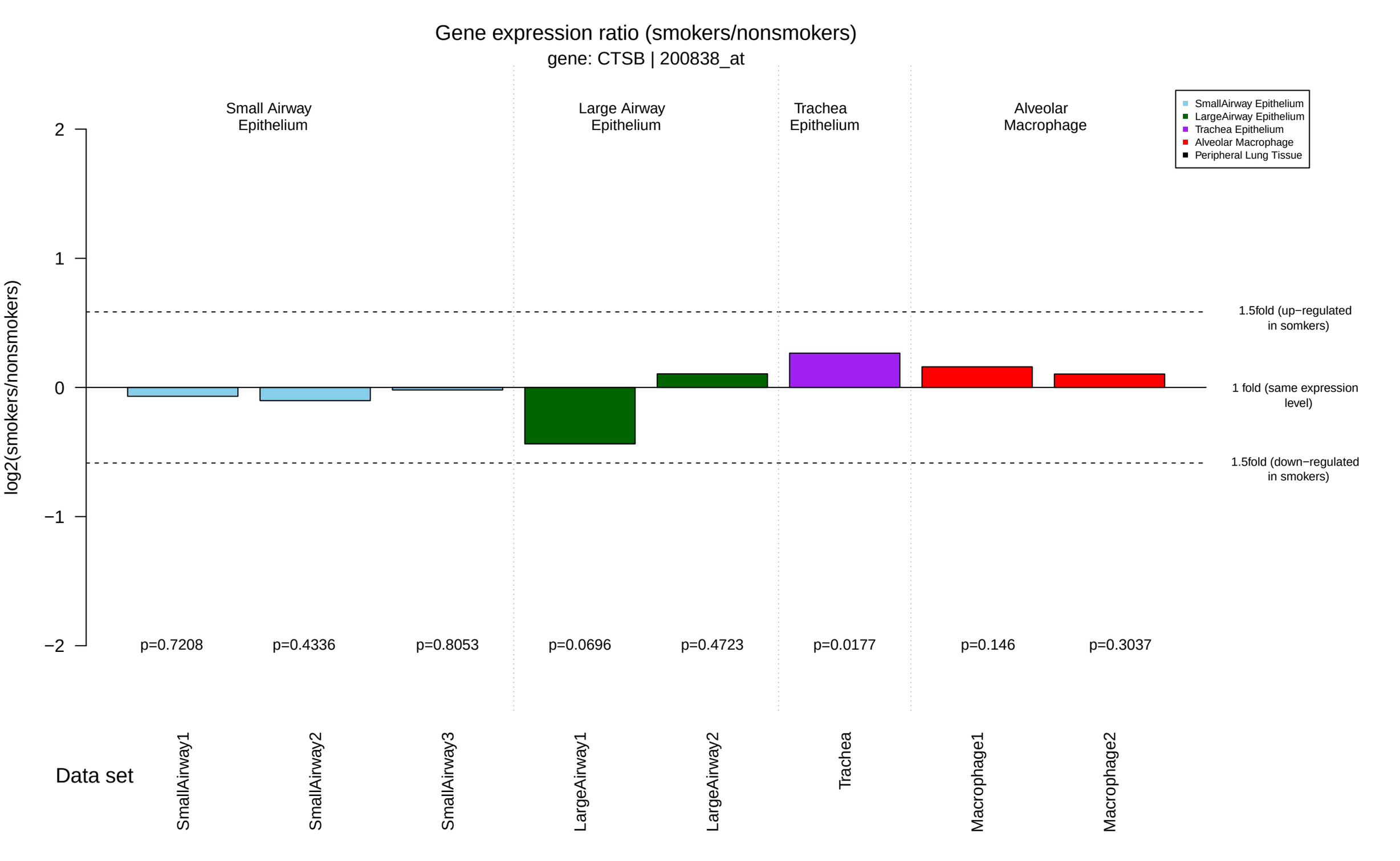
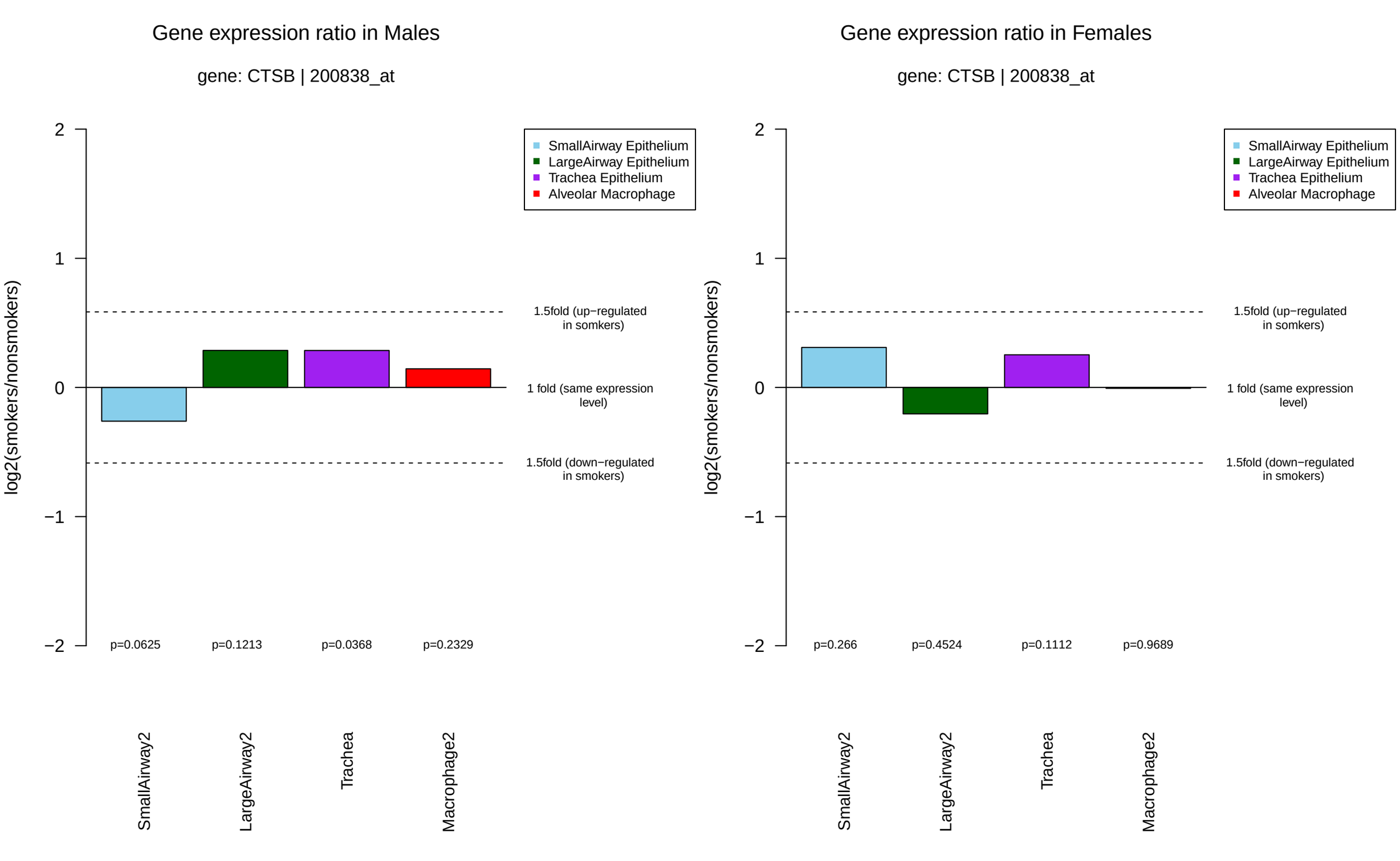
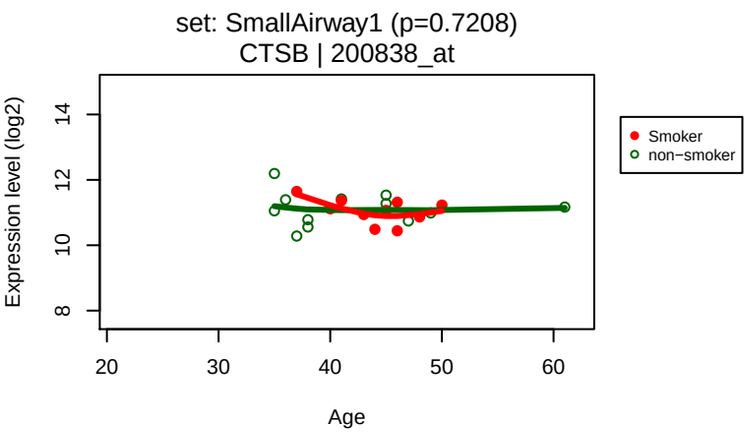
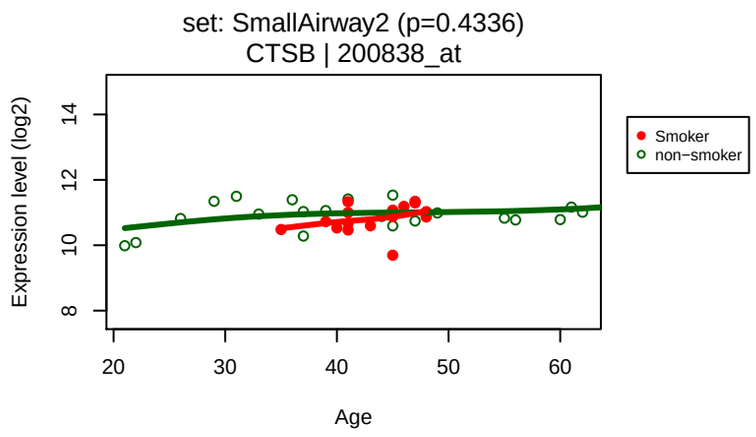
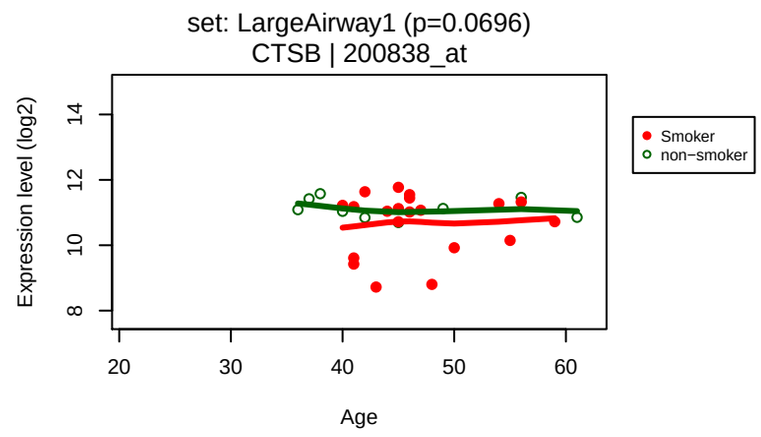

Gene expression of smokers and nonsmokers
gene: CTSB | 200838_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7208
p=0.4336
p=0.8053
p=0.0696
p=0.4723
p=0.0177
p=0.146
p=0.3037
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

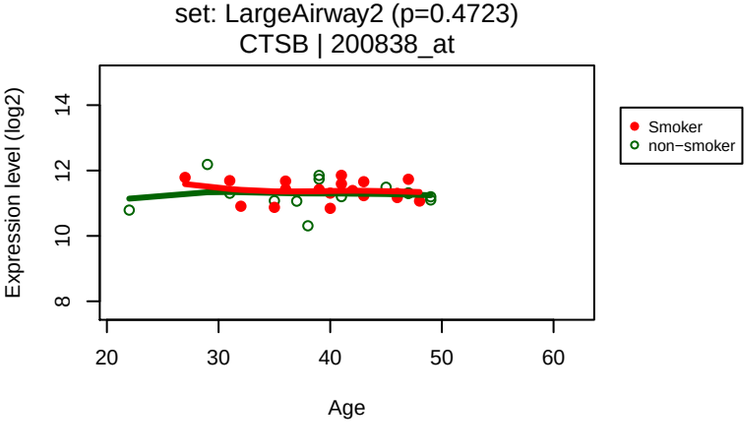

Gene expression ratio (smokers/nonsmokers)
gene: CTSB | 200838_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7208
p=0.4336
p=0.8053
p=0.0696
p=0.4723
p=0.0177
p=0.146
p=0.3037
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

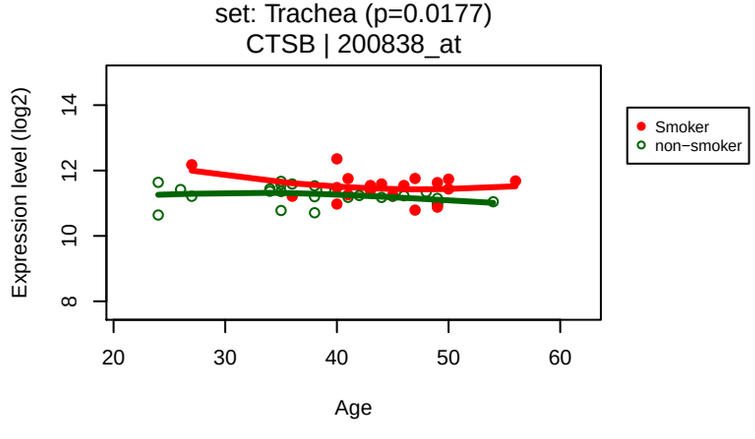

Gene expression ratio in Males
gene: CTSB | 200838_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0625
p=0.1213
p=0.0368
p=0.2329
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSB | 200838_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.266
p=0.4524
p=0.1112
p=0.9689
SmallAirway2
LargeAirway2
Trachea
Macrophage2

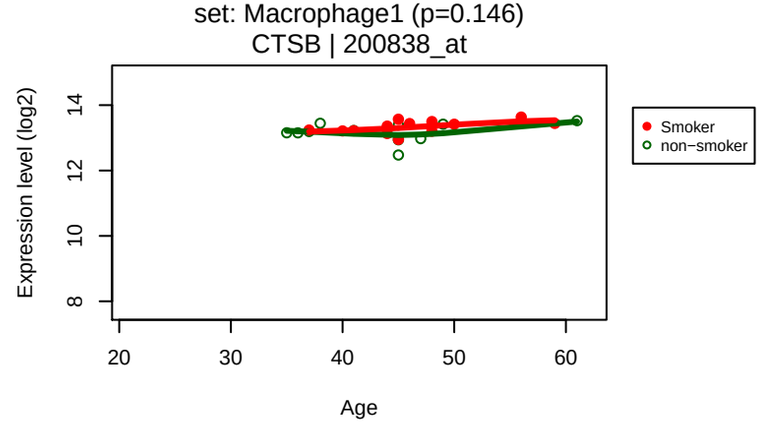

set: SmallAirway1 (p=0.7208)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker

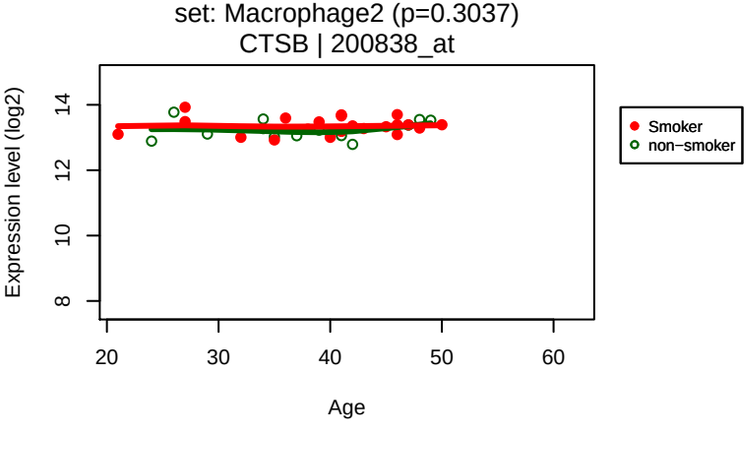

set: SmallAirway2 (p=0.4336)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker

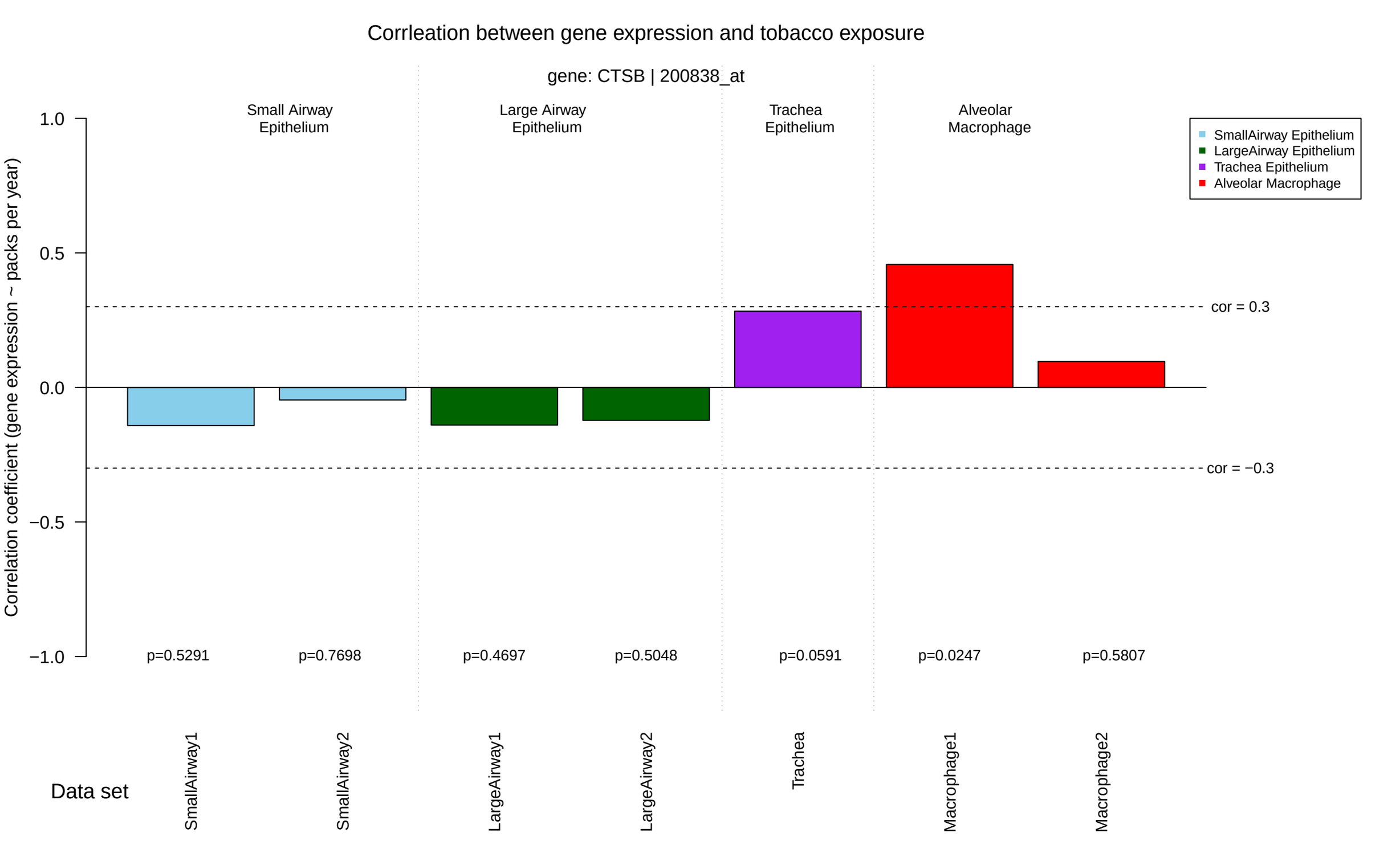

set: LargeAirway1 (p=0.0696)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4723)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0177)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.146)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3037)
CTSB | 200838_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSB | 200838_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5291
p=0.7698
p=0.4697
p=0.5048
p=0.0591
p=0.0247
p=0.5807
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2